STRATEGY OVERVIEW · NQ / MNQ

Prior VWAP Trap

E-mini Nasdaq-100 futures · 1-minute bars · Chicago time · long and short · at most one trade a day in the cash session · flat by 15:00 CT · one contract

1 NQ, whole backtest	Net	PF	Trades / yr	Win rate	Average	Max DD	Closed DD
MultiCharts · all trades	$290,640	1.25	192	46.3%	$154.84	$41,195	—
MultiCharts · longs	$159,545	1.27	100	—	$163.47	—	—
MultiCharts · shorts	$131,095	1.22	92	—	$145.50	—	—
Research simulation · all trades	**$296,937**	**1.25**	**192**	**46.3%**	**$157.95**	—	**$36,925**

MultiCharts backtest on a 1-minute @NQ chart (dxFeed continuous, unadjusted), 2017-01-17 → 2026-10-27, 1,877 trades (976 long / 901 short). Research = the same rule simulated in Python on the same 1-minute data, 2017-01-03 → 2026-10-22, 1,880 trades; on the window both cover it gives 1,875 trades against MultiCharts' 1,877. $2.50 commission per contract per side and $5 slippage per order ($15 a round trip). 1 NQ = $20 a point (1 MNQ = one tenth). Max DD counts closed equity plus each trade's worst open loss; Closed DD counts closed trades only. Long / short PF and averages are arithmetic from the MultiCharts report's gross profit and loss.

Equity path and drawdown



Research simulation, closed trades, 1 NQ. The dashed line marks 2023-01-01: everything after it was run once, after the rule had been frozen on 2017–2022.

The trade story

Yesterday's cash-session VWAP is the price yesterday's participants paid on average. After a down or flat day the average participant was a seller; when price closes back above that VWAP today, that crowd is under water, and its covering carries the move. After an up or flat day the mirror traps yesterday's buyers below it. If yesterday already moved more than half a normal day's range in the trade's direction, the crowd is already on the trade's side and there is nobody left to trap, so that side is skipped. The trade is held to 15:00 CT behind a stop that trails at half a normal day's range. This is the strategy's own reading of who pays; it has not been measured, and no published paper describes this exact effect.

Rules

Session	The 08:31–15:00 CT cash session of the Globex date (bars stamped at their close; 17:00 CT starts the next date).
Level	PVWAP = yesterday's cash-session VWAP: sum(close × volume) ÷ sum(volume) over yesterday's cash bars, moved onto today's contract on roll days.
Day filter	ADR10 = mean cash-session high–low range of the previous 10 sessions. PrevMove = (yesterday's close − the close before) ÷ ADR10. Long side allowed if PrevMove ≤ +0.5; short side if PrevMove ≥ −0.5. Today and yesterday need at least 200 cash bars.
Entry	On 1-minute closes, filled at the next bar's open; the first trigger wins; one trade a day. **Cross:** the session opened on the far side of PVWAP and a bar from 08:35 closes through it. **Reclaim:** the session opened beyond PVWAP, a close came back across it, then a close goes beyond again. **Hold:** the session opened beyond PVWAP and no close came back across through 10:00; entry at the 10:01 open. No entries in the last 180 minutes of the session (after 12:00 on a normal day).
Stop	0.5 × ADR10 from the entry, trailed 0.5 × ADR10 behind the best high (long) or low (short) since the fill; never loosened. The stop reaches the entry price once the trade has moved 1R (0.5 × ADR10) in its favour.
Exit	At the open of the 15:00 CT bar; on an early-close day, the close of the last cash bar. No target.
Parameters	PrevMove limit 0.5 · first cross/reclaim bar 08:35 · hold check 10:00 · no entries in the last 180 min · exit 15:00 · stop and trail 0.5 × ADR · ADR length 10 · minimum 200 cash bars · one side a day. All are inputs in the code; the values are the first written ones.

By year, research simulation, 1 NQ

Year	Long trades	Long net	Short trades	Short net	Total
2017	101	$7,951	97	−$4,210	**$3,741**
2018	100	$6,714	83	$8,835	**$15,549**
2019	98	$10,033	99	−$3,326	**$6,707**
2020	97	$31,888	87	$23,979	**$55,867**
2021	98	$16,373	94	$10,027	**$26,400**
2022	96	$9,515	87	$54,551	**$64,066**
2023	101	$42,522	89	−$32,335	**$10,187**
2024	100	$26,334	92	$10,277	**$36,611**
2025	108	−$6,816	93	$14,987	**$8,171**
2026	78	$20,578	82	$49,062	**$69,640**

By exit year; 2026 to 2026-10-22. Every year positive in total. Positive months: 69 of 118.

Worst drawdowns, research simulation, closed trades, 1 NQ

Depth	Peak	Low	Recovered	Days
$36,925	2022-06-09	2023-08-11	2024-08-06	788
$36,682	2025-04-02	2025-10-21	2026-03-02	334
$25,715	2026-05-13	2026-06-04	2026-06-23	41
$23,487	2026-08-14	2026-10-08	not yet	69+

On 2026-10-22 the equity was $4,123 below its peak. Largest single win $14,463, largest single loss −$6,528. The MultiCharts run's max drawdown including open-trade loss: $41,195.

Why it should work

- **Written down before it ran:** the first written version was taken forward untouched; no parameter was tuned and no filter was added.
- **Development 2017–2022:** 1,137 trades, $172,328, PF 1.319, t-statistic 3.13, six of six years positive.
- **Stable around its parameters:** every parameter moved ±10% (16 cases): all net positive, PF 1.23–1.35.
- **Run once on later data:** 2023-01-01 → 2026-10-23: 743 trades, $124,608, PF 1.197, every year positive.
- **Both sides pay** over the whole period: longs PF 1.27, shorts PF 1.22 in MultiCharts.
- **2022, a falling year:** NQ futures fell 32.5% (continuous contract, 2021 to 2022 cash close); the research run made $64,066 (shorts $54,551, longs $9,515).

Risks and caveats

- **The edge thinned after 2023:** PF 1.319 → 1.197, t-statistic 3.13 → 1.85, win rate 47.3% → 44.7%. Dollars per trade held ($151.56 → $167.71); the test's reading is that NQ's daily range roughly doubled. The average loss grew faster than the average win (loss $902 → $1,535; win $1,325 → $2,276).
- **Long flat periods:** the worst drawdown ran from 2022-06-09 to 2023-08-11 and took until 2024-08-06 to recover (788 days).
- **The sides take turns:** shorts lost −$32,335 in 2023; longs lost −$6,816 in 2025.
- **No published effect:** the "trapped crowd" is the strategy's reading, not a measured flow.
- **Roll handling:** PVWAP is moved onto the new contract with a list of detected price jumps. The list holds three reopen gaps that are not rolls (2024-12-12, 2026-03-08, 2026-09-13) and misses the June 2024 roll; it is kept as is because the research used it.
- **Fills:** in MultiCharts the stop starts on the bar after the fill; the research also stopped trades inside the fill bar. On the same window MultiCharts made $290,640 against the research's $298,294.

Robustness grade

Split at 2023-01-01, 1 NQ	Prior VWAP Trap C
Trades in / out of sample	1,132 / 745
Expected value per trade, in → out	$152.11 → $158.99
Kept out of sample · P(edge ≤ 0)	105% · 0.04
95% interval of the out-of-sample edge per trade	−$16.87 to $336.48
Out-of-sample net / max DD (closed)	$118,450 / $36,855
Return ÷ drawdown, out of sample	3.21
Monte-Carlo max DD, median / 75th / 95th / 99th	$42,005 / $51,430 / $70,966 / $88,475

Graded 2026-10-29 on the MultiCharts List of Trades (1,877 trades, 1 NQ, costs included), split by entry date at 2023-01-01; risk unit = the in-sample average loss ($909.73). Expected value from a 10,000-draw bootstrap; drawdowns from a 20,000-path Monte Carlo of the trade sequence (closed equity). Grade scale S–D combines edge kept out of sample, confidence it is above zero, and return ÷ drawdown. C rather than B: the out-of-sample return ÷ drawdown is 3.21 (B needs 4) and the out-of-sample interval reaches below zero (B needs it above zero); the edge itself was kept in full.

What has been done to it

#	Step	Result
1	Rule written down before any test	The version shown here; never changed afterwards.
2	Research run on 2017-01-01 → 2022-12-31	1,137 trades (590 long / 547 short), $172,328, PF 1.319, t 3.13, max drawdown $32,729 (marked to market).

3	Robustness: each parameter moved ±10%	16 cases, all net positive, PF 1.23–1.35.
4	Rule frozen, then run once on 2023-01-01 → 2026-10-23 against pass checks set in advance	743 trades, $124,608, PF 1.197, t 1.85, max drawdown $40,776. Passed: net above zero, at least 350 trades, time in market under 15%, and average trade, net per year and net per year ÷ drawdown each kept within 25% of development. Failed one: the profit factor's part above 1.0 fell 38% (limit 25%; it needed 1.239).
5	Translated to MultiCharts (PowerLanguage) and checked against the research	1,877 trades against the research's 1,875 on the same window; $290,640 against $298,294 (the fill-bar stop difference above).

Kill test — what would prove it wrong

Watch	From the data, 1 NQ	Warning sign
Drawdown	Worst so far $41,195 with open loss, $37,275 on closed trades; out of sample (2023 →) $36,855 closed. Monte Carlo of the trade sequence (closed equity): median $42,005, 75th percentile $51,430, 95th $70,966, 99th $88,475.	Beyond the 75th percentile ($51,430).
Edge per trade	Bootstrap expected value: $152.11 a trade before 2023 (95% interval $59.57 to $247.82) → $158.99 from 2023 (−$16.87 to $336.48); probability the 2023+ edge is zero or less 0.04.	Expected value per trade falling toward zero.
Win rate vs the win rate it needs	46.3% vs 40.9% (average win $1,689, average loss $1,168)	Win rate sliding toward the needed level.
Trade count	≈ 192 a year: 1,132 trades before 2023, 745 from 2023	Materially different: data or session setup changed.
Falling markets	2022: +$63,825 while NQ fell 32.5%	A falling year in which it loses.

From the MultiCharts List of Trades (1,877 trades, 2017-01-17 → 2026-10-26, 1 NQ, costs included), split at 2023-01-01. Expected value from a 10,000-draw bootstrap; drawdowns from a 20,000-path Monte Carlo of the trade sequence (closed equity, per contract). Needed win rate = average loss ÷ (average win + average loss).

Source: MultiCharts backtest and a Python research simulation of the same rule on dxFeed NQ 1-minute data; rules from the strategy code (PowerLanguage, sent with this document). Backtests, not account results. Prepared 2026-10-29.